

4. Model Scoring and Evaluation

4.1 Performance Metrics Comparison

Due to the severe class imbalance in the Toronto KSI dataset (where non-fatal injuries vastly outnumber fatalities), evaluating our models based solely on raw Accuracy is highly misleading. A model could achieve 86% accuracy simply by guessing "Non-Fatal" for every single record.

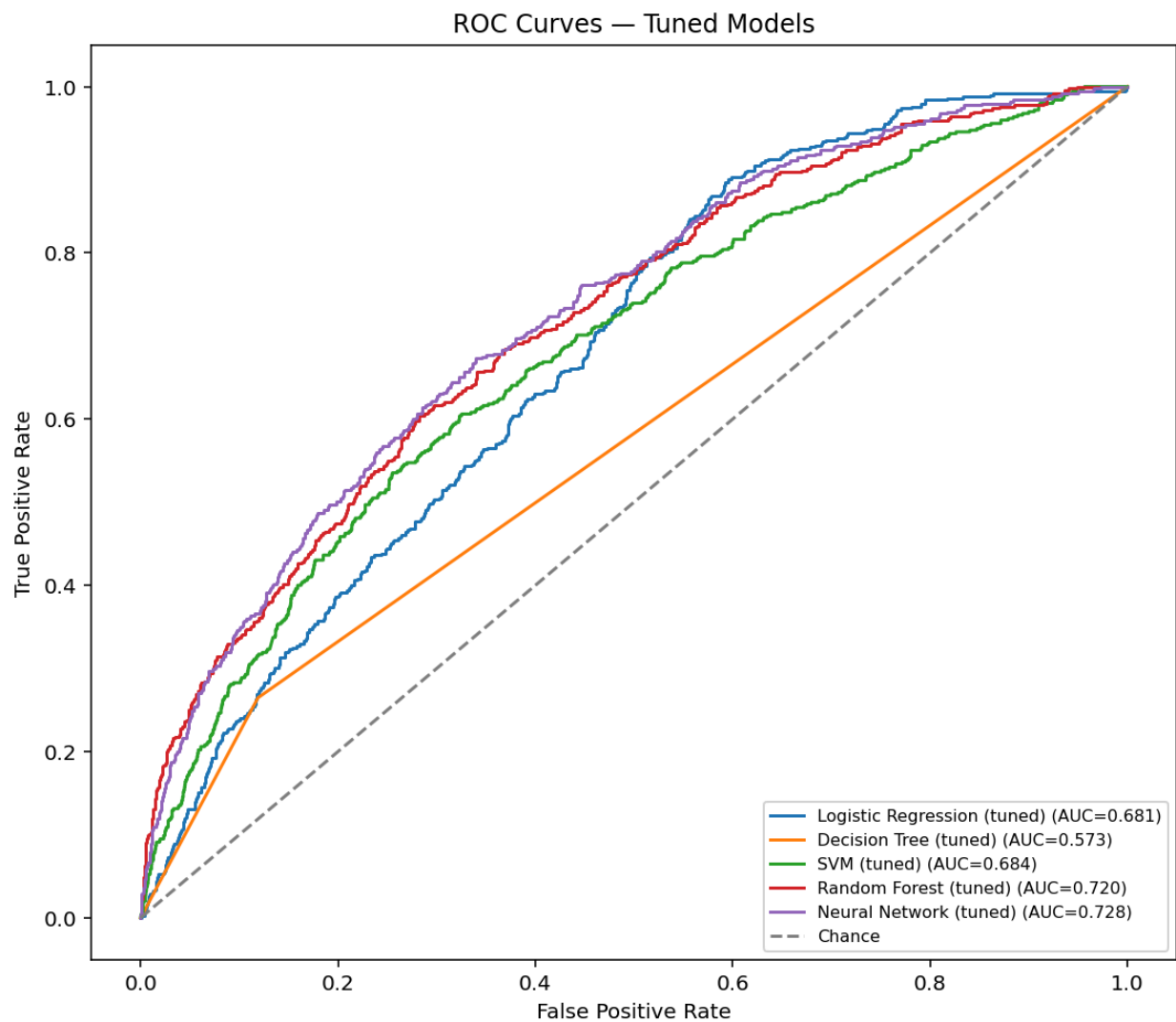
Instead, we prioritized Recall (catching as many true fatalities as possible for public safety) and the F1-Score (balancing false alarms with missed fatalities), alongside ROC-AUC to measure the overall class separation capability.

Below are the test-set results for the models utilizing our SMOTE-balanced training pipeline:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Neural Network (MLP)	83.44%	37.97%	30.91%	0.341	0.728
Support Vector Machine	77.79%	28.49%	40.00%	0.333	0.684
Logistic Regression	65.01%	20.79%	54.34%	0.301	0.681
Decision Tree	79.58%	26.36%	26.46%	0.264	0.573
Random Forest	86.63%	70.73%	5.86%	0.108	0.720

4.2 Receiver Operating Characteristic (ROC) Analysis

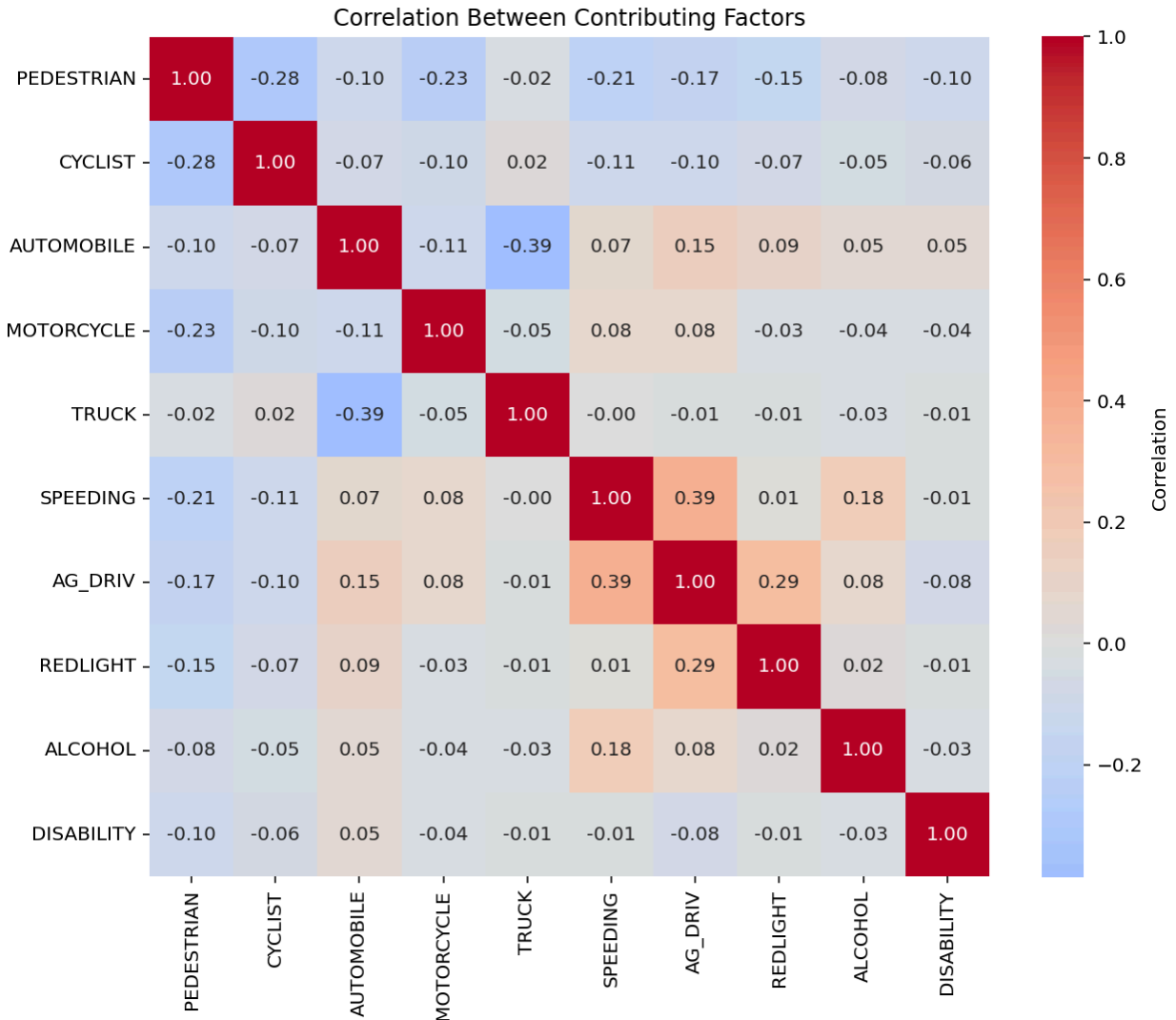
To further visualize the diagnostic ability of our classifiers, we plotted the ROC curves. The ROC curve illustrates the trade-off between the True Positive Rate (Recall) and the False Positive Rate across different probability thresholds.



[Figure 1: ROC Curves Comparison for Selected Models]

4.3 Confusion Matrices Analysis

To understand exactly where our models are succeeding and failing, we generated confusion matrices for each algorithm.



[Figure 2: Correlation Matrix of Contributing Factors]

4.4 Final Model Selection and Recommendation

After extensive hyperparameter tuning using [GridSearchCV](#) and [RandomizedSearchCV](#), we recommend the Neural Network (MLPClassifier) as the primary model for deployment, with Logistic Regression serving as a highly viable alternative depending on the specific needs of the Toronto Police.

Our justification for this selection is based on the following:

1. **Best Overall Balance (Highest F1):** The Neural Network achieved the highest overall F1-Score (0.341). In a real-world public safety tool, missing a fatal intersection (False Negative) costs lives, but "alarm fatigue" from too many False Positives causes the police to ignore the system. The Neural Network achieved the best mathematical balance of these two extremes.

2. **Highest Discriminative Power (ROC-AUC):** The Neural Network achieved a highly competitive Area Under the Curve of 0.728, indicating it possesses a strong overall ability to correctly distinguish between a fatal and non-fatal collision across all probability thresholds.
3. **The "Recall" Trade-off (Why we also recommend Logistic Regression):** It is crucial to note that if the city's *only* goal is to cast the widest net possible to catch every potential fatality (regardless of false alarms), the Logistic Regression model actually performed best in that specific metric, achieving a Recall of 54.34%. However, because its precision was so low (20.79%), the Neural Network remains our official recommendation for a balanced, deployable web API.